

Programme	B. Sc. STATISTICS				
Course Code	STA 7 CJ 404 (P)				
Course Title	ADVANCED SAMPLING METHODS AND DESIGN OF EXPERIMENTS				
Type of Course	Major				
Semester	VII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practicum per week	Total Hours
	4	3	-	2	75
Pre-requisites	Knowledge about sampling procedures and various sampling methods, linear estimation and analysis of variance				
Course Summary	Understand PPS sampling, ratio and regression sampling methods. Identify various factorial design experiments.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Estimate the population parameters concerning the study variables under auxiliary information (Ratio and regression methods).	An	C	Instructor-created exams
CO2	Discuss probability proportional to size (PPS) sampling strategies.	U	F	Home Assignments
CO3	Explain the concepts of ordered and unordered estimators and its properties.	Ap	C	Instructor-created exams
CO4	Discuss the multi stage and multiphase sampling.	U	F	Seminar Presentation
CO5	Apply incomplete block designs and balanced incomplete block designs.	Ap	C	Practical Assignment
CO6	Explain factorial experiments, total confounding and partial confounding.	Ap	C	Group Tutorial Work
CO7	Differentiate between strip plot and split plot designs.	An	P	Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I		Open-Ended	30	
		Census and Sampling-Basic concepts, probability sampling and non-probability sampling, simple random sampling with and without replacement, Systematic sampling- linear and circular systematic sampling, Stratification and stratified random sampling, Complete Block Designs. Problems using R/Python		
II		Cluster , Ratio and Regression Sampling	12	20
	1	Cluster sampling with equal and unequal clusters		
	2	Estimation of mean and variance, relative efficiency, optimum cluster size, varying probability cluster sampling		
	3	Ratio method of estimation-estimation of ratio, mean and total.		
	4	Bias and relative bias of ratio estimator. Mean square error of ratio estimator. Unbiased ratio type estimator		
	5	Regression methods of estimation		
	6	Comparison of ratio and regression estimators with simple mean per unit method. Ratio and regression method of estimation in stratified population		
III		Varying probability sampling	10	15
	7	pps sampling with and without replacements		
	8	Des- Raj ordered estimators-Murthy's unordered estimator		
	9	Horvitz-Thompson estimators, Yates and Grundy forms of variance and its estimators		
	10	Zen-Midzuno scheme of sampling, π PS sampling		
	11	Multi stage and multiphase sampling		
IV		Incomplete Block Designs	11	20
	12	Balanced Incomplete Block designs		
	13	Construction of BIB Designs, Analysis with recovery of inter-block information and intra-block information		
	14	Partially balanced incomplete block designs		
	15	Analysis of partially balanced incomplete block designs with two associate classes		
	16	Youden square design		
	17	Lattice designs		
V		Factorial Designs	12	15
	18	Basic definitions and principles - Analysis of 2^n factorial experiments		
	19	Total confounding of 2^n designs in 2^n blocks. Partial confounding in 2^n blocks		
	20	3^n factorial designs		
	21	Fractional factorial designs		
	22	Concepts of Split plot design and strip plot design		

Text Books

1. **Cochran W.G. (1992)**: Sampling Techniques, Wiley Eastern, New York.
2. **D. Singh and F.S. Chowdhary (1986)**: Theory and Analysis of Sample Survey Design, Wiley Eastern (New Age International), New Delhi.
3. **Montgomery D C (2001)**. Design and Analysis of Experiments, John Wiley.
4. **Das M N and Giri N C (1979)**. Design and Analysis of Experiments, second edition, Wiley.

References

1. **P.V.Sukhatme et.al. (1984)**: Sampling Theory of Surveys with Applications. IOWA State University Press, USA.
2. **Des Raj (1976)**: Sampling Theory. McGraw Hill
3. **Mukhopadhyay. P. (1999)**. Theory and Methods of Survey Sampling. Prentice-Hall India, New-Delhi.
4. **Chakrabarti, M.C. (1964)**. Design of experiments, ISI, Calcutta.
5. **Hinkleman and Kempthorne C (1994)**. Design and Analysis of Experiments Volume I, John Wiley.